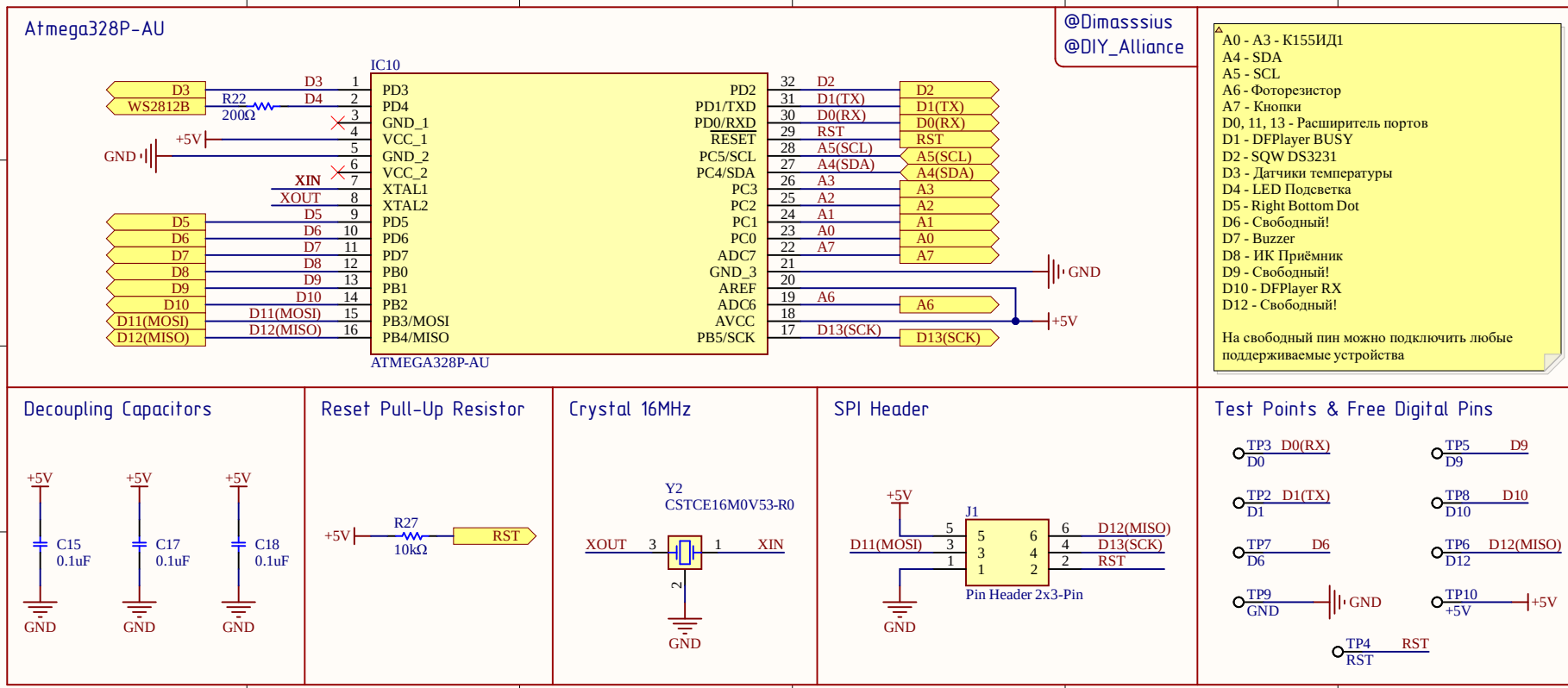
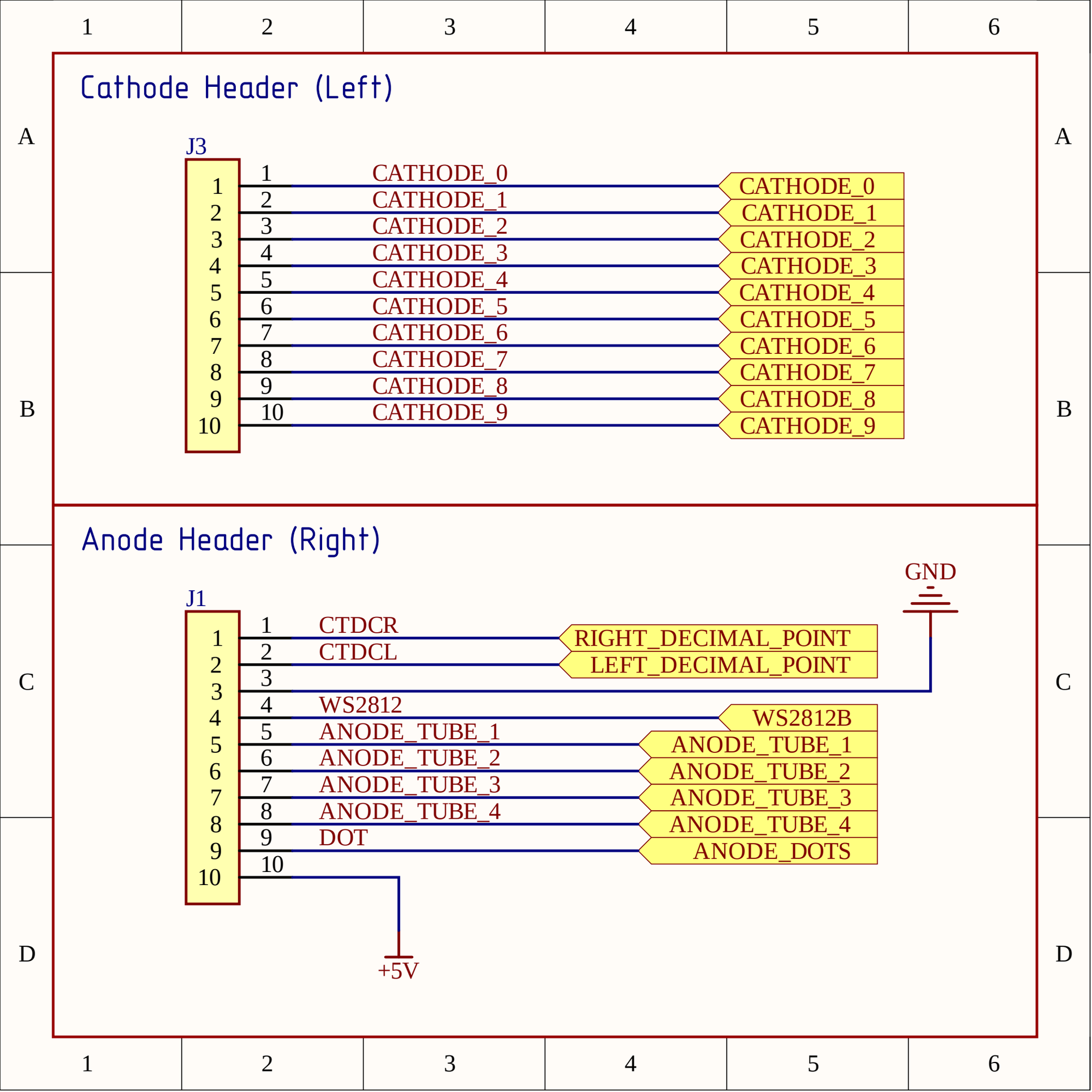
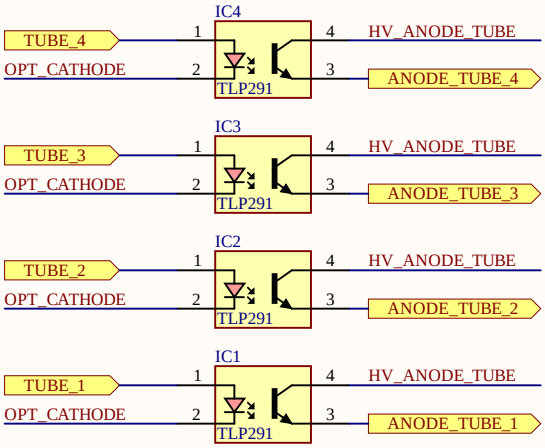
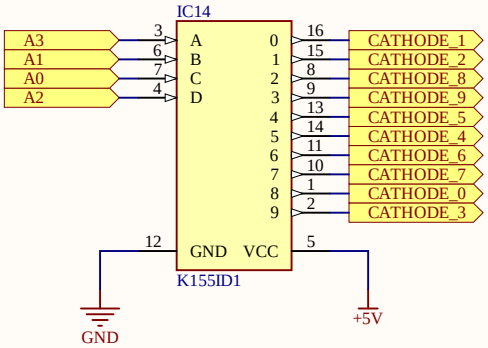

Atmega328P-AU
@Dimasssius
@DIY_Alliance
IC10
D3
WS2812B
R22
200Ω
D3
D4
+5V
GND
XIN
XOUT
D5
D6
D7
D8
D9
D10
D11(MOSI)
D12(MISO)
PD3
PD4
GND_1
VCC_1
GND_2
VCC_2
XTAL1
XTAL2
PD5
PD6
PD7
PB0
PB1
PB2
PB3/MOSI
PB4/MISO
PD2
PD1/TXD
PD0/RXD
RESET
PC5/SCL
PC4/SDA
PC3
PC2
PC1
PC0
ADC7
GND_3
AREF
ADC6
AVCC
PB5/SCK
D2
D1(TX)
D0(RX)
RST
A5(SCL)
A4(SDA)
A3
A2
A1
A0
A7
A6
D13(SCK)
ATMEGA328P-AU
A0 - A3 - К155ИД1
A4 - SDA
A5 - SCL
A6 - Фоторезистор
A7 - Кнопки
D0, 11, 13 - Расширитель портов
D1 - DFPlayer BUSY
D2 - SQW DS3231
D3 - Датчики температуры
D4 - LED Подсветка
D5 - Right Bottom Dot
D6 - Свободный!
D7 - Buzzer
D8 - ИК Приёмник
D9 - Свободный!
D10 - DFPlayer RX
D12 - Свободный!
На свободный пин можно подключить любые поддерживаемые устройства
Decoupling Capacitors
C15
0.1uF
C17
0.1uF
C18
0.1uF
Reset Pull-Up Resistor
R27
10kΩ
Crystal 16MHz
Y2
CSTCE16M0V53-R0
SPI Header
J1
Pin Header 2x3-Pin
Test Points & Free Digital Pins
TP3 D0(RX)
TP5 D9
TP2 D1(TX)
TP8 D10
TP7 D6
TP6 D12(MISO)
TP9 GND
TP10 +5V
TP4 RST


Cathode Header (Left)
J3
1
2
3
4
5
6
7
8
9
10
CATHODE_0
CATHODE_1
CATHODE_2
CATHODE_3
CATHODE_4
CATHODE_5
CATHODE_6
CATHODE_7
CATHODE_8
CATHODE_9
Anode Header (Right)
J1
CTDCR
CTDCL
WS2812
ANODE_TUBE_1
ANODE_TUBE_2
ANODE_TUBE_3
ANODE_TUBE_4
DOT
RIGHT_DECIMAL_POINT
LEFT_DECIMAL_POINT
WS2812B
ANODE_DOTS
GND
+5V

Anode Tubes Drivers



Decoder - K155ID1



Anode Tubes Resistor

Данным резистором можно регулировать яркость свечения индикаторов и разделительных точек, чем меньше сопротивление, тем выше яркость (и выше ток, что влияет на срок жизни индикаторов)

Варианты:
1. 33 кОм
2. 10 кОм (рекомендуемый)
3. 4.7 кОм
4. 3.3 кОм



Optoisolators Cathode Resistor

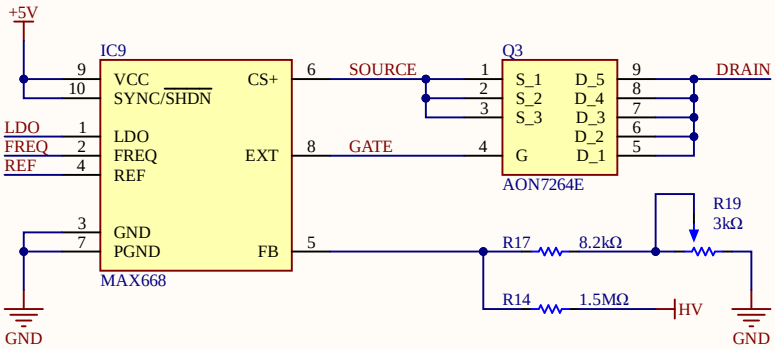


Test Point - High Voltage

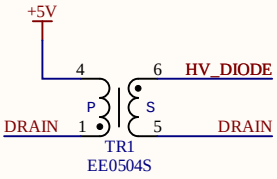


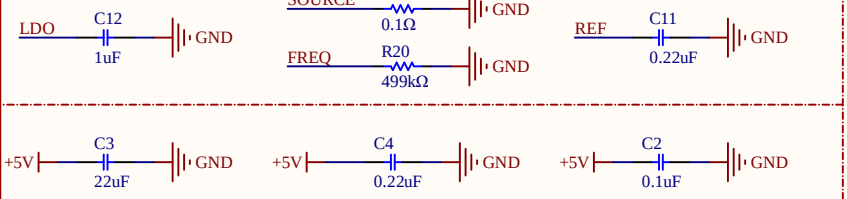
@Dimasssius
@DIY_Alliance

High Voltage Generator

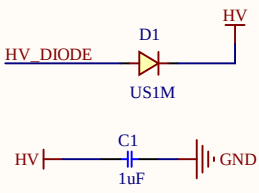


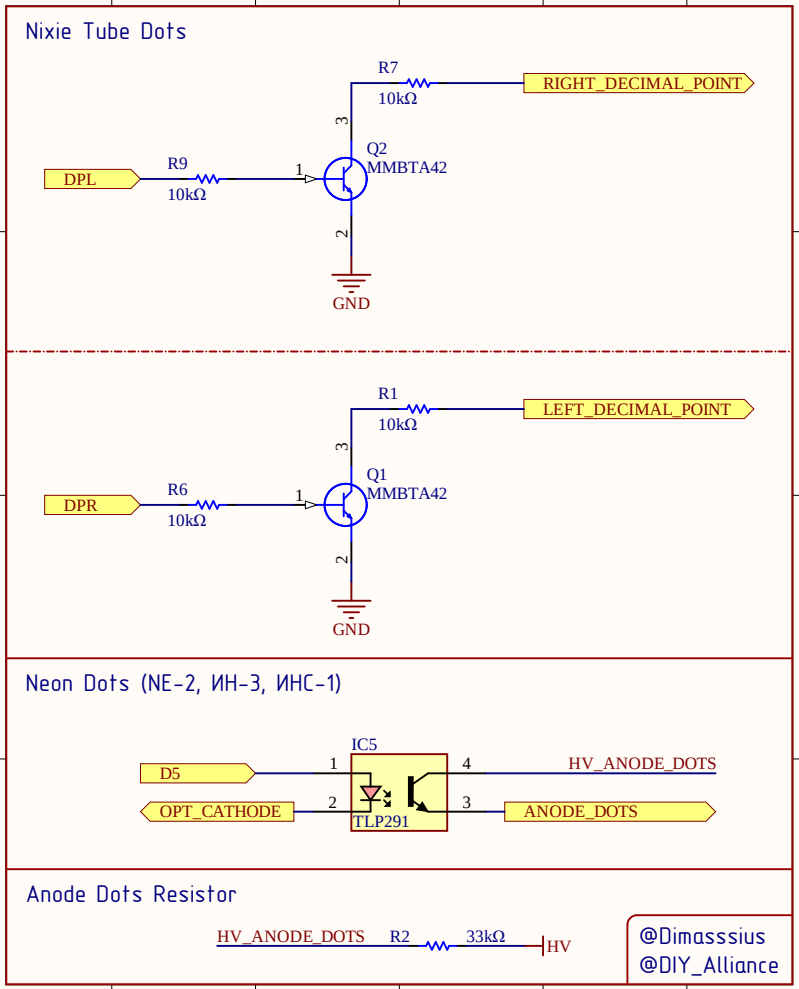
Transformer 1:10

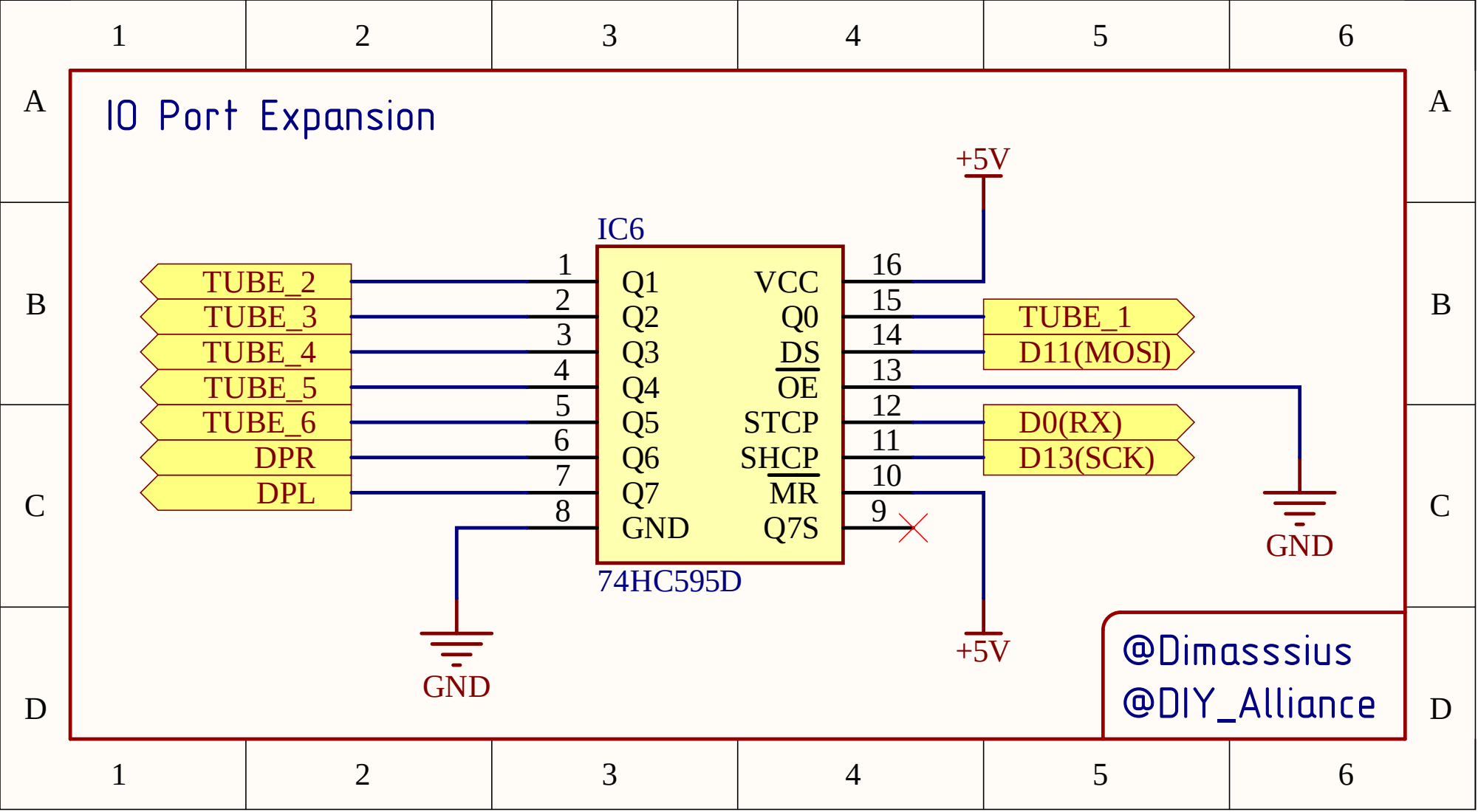


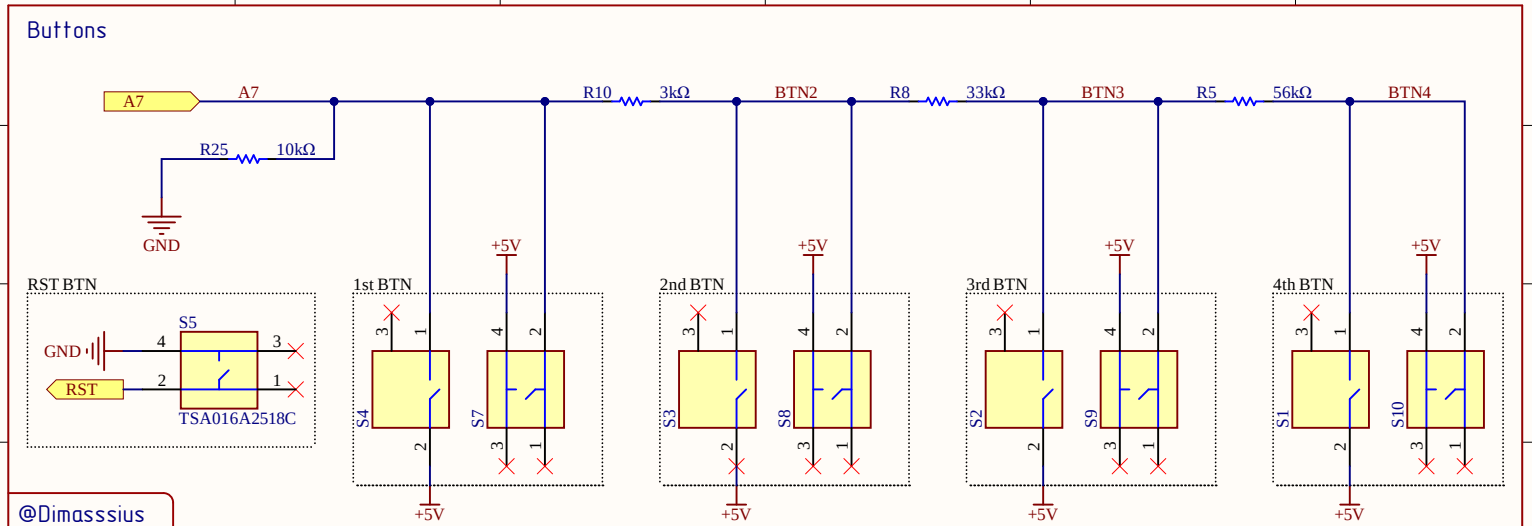


+180V

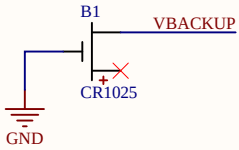



Nixie Tube Dots
R7
10kΩ
RIGHT_DECIMAL_POINT
DPL
R9
10kΩ
Q2
MMBTA42
GND
R1
10kΩ
LEFT_DECIMAL_POINT
DPR
R6
10kΩ
Q1
MMBTA42
GND
Neon Dots (NE-2, ИН-3, ИНС-1)
IC5
D5
OPT_CATHODE
TLP291
HV_ANODE_DOTS
ANODE_DOTS
Anode Dots Resistor
HV_ANODE_DOTS
R2
33kΩ
HV
@Dimasssius
@DIY_Alliance

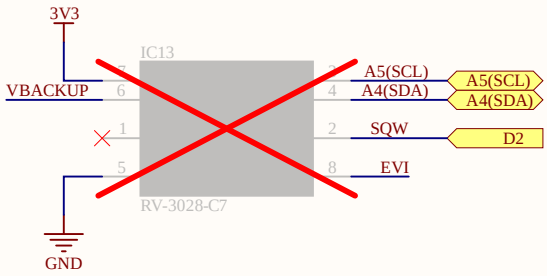

IO Port Expansion
IC6
74HC595D
TUBE_2
TUBE_3
TUBE_4
TUBE_5
TUBE_6
DPR
DPL
1 Q1
2 Q2
3 Q3
4 Q4
5 Q5
6 Q6
7 Q7
8 GND
16 VCC
15 Q0
14 DS
13 OE
12 STCP
11 SHCP
10 MR
9 Q7S
+5V
+5V
TUBE_1
D11(MOSI)
D0(RX)
D13(SCK)
GND
GND
@Dimasssius
@DIY_Alliance


Buttons
A7
A7
R25
10kΩ
GND
R10
3kΩ
BTN2
R8
33kΩ
BTN3
R5
56kΩ
BTN4
RST BTN
GND
RST
S5
TSA016A2518C
1st BTN
2nd BTN
3rd BTN
4th BTN
+5V
S4
S7
S3
S8
S2
S9
S1
S10
@Dimasssius
@DIY_Alliance

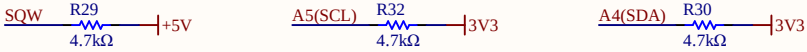
Battery Holder



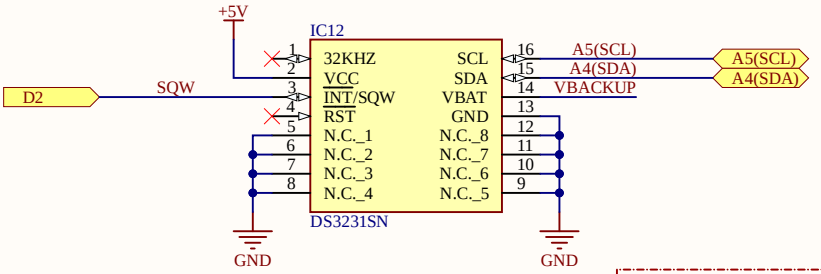
RV-3028-C7 - Variant #1



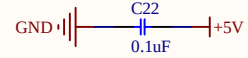
Pull-Ups

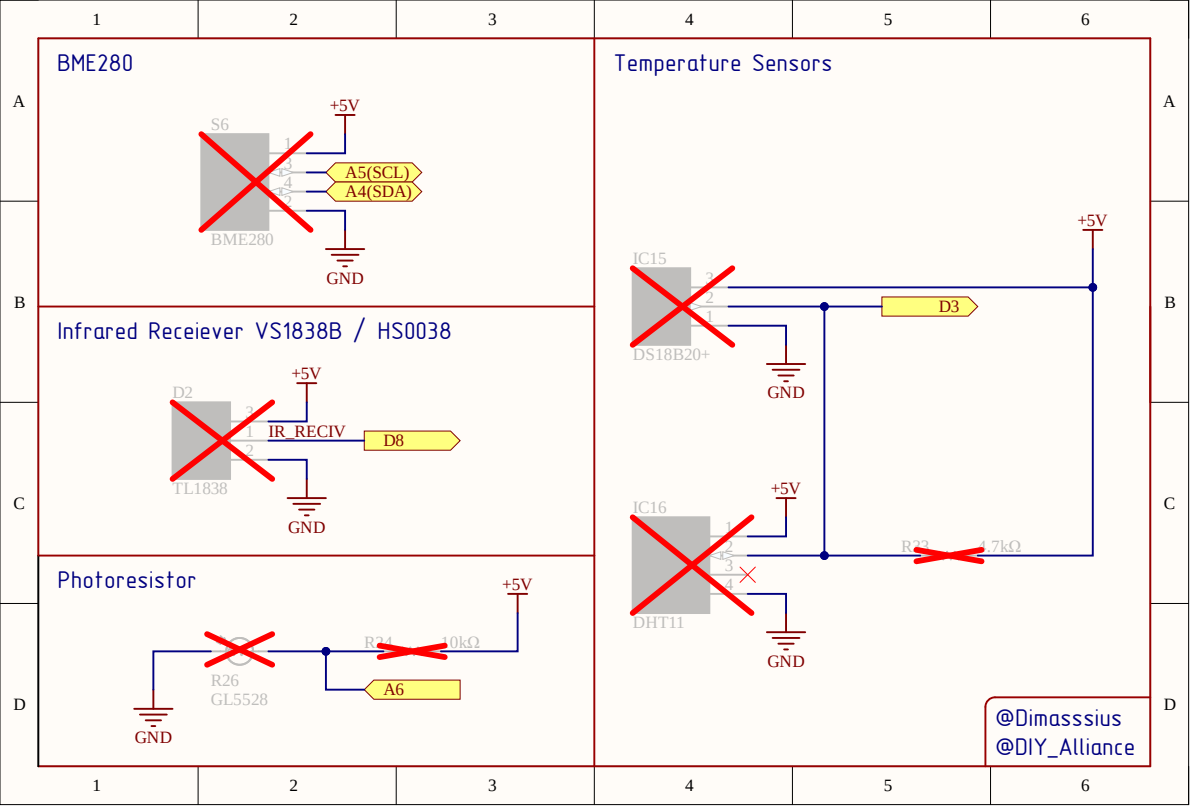


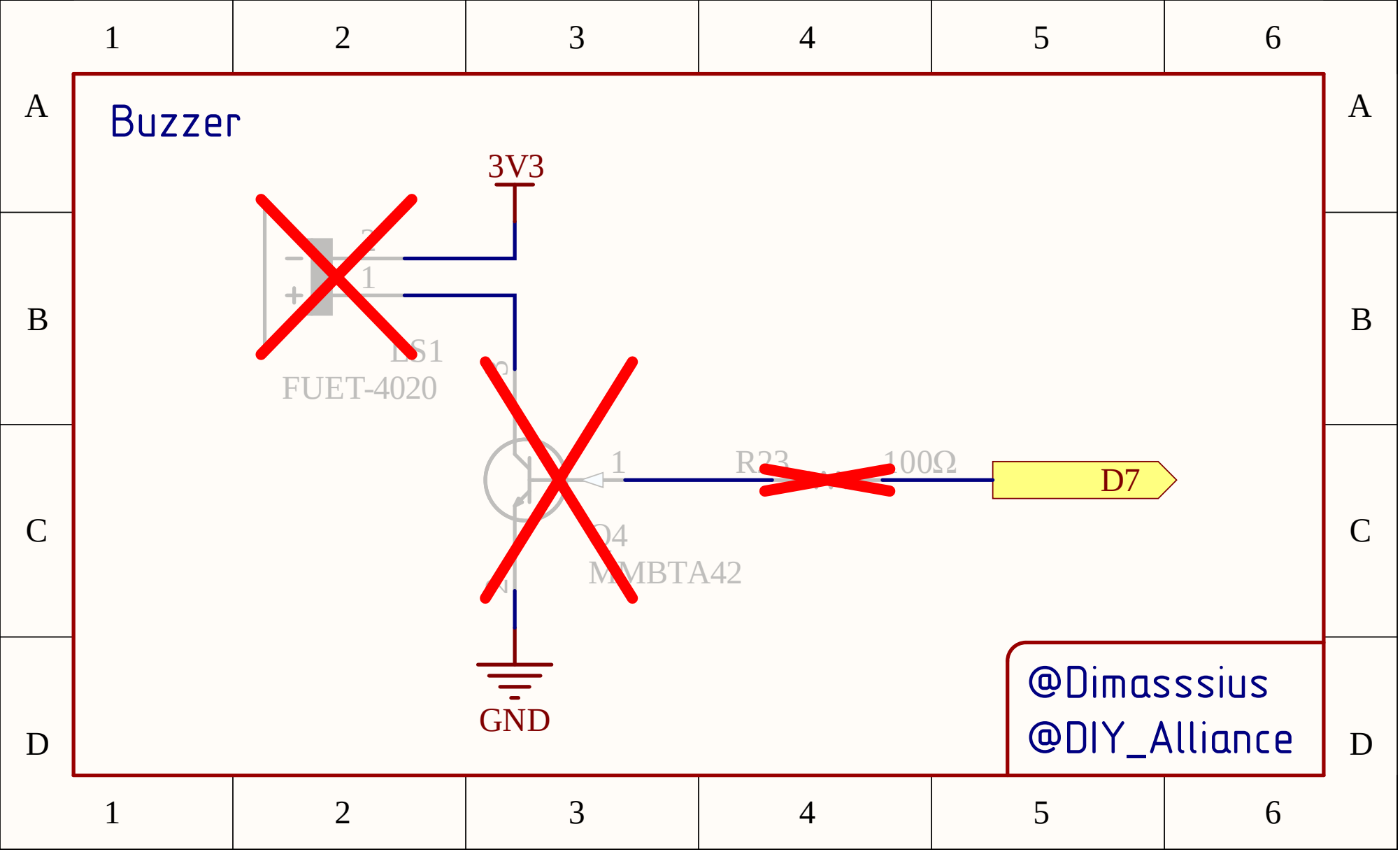
DS3231 - Variant #2

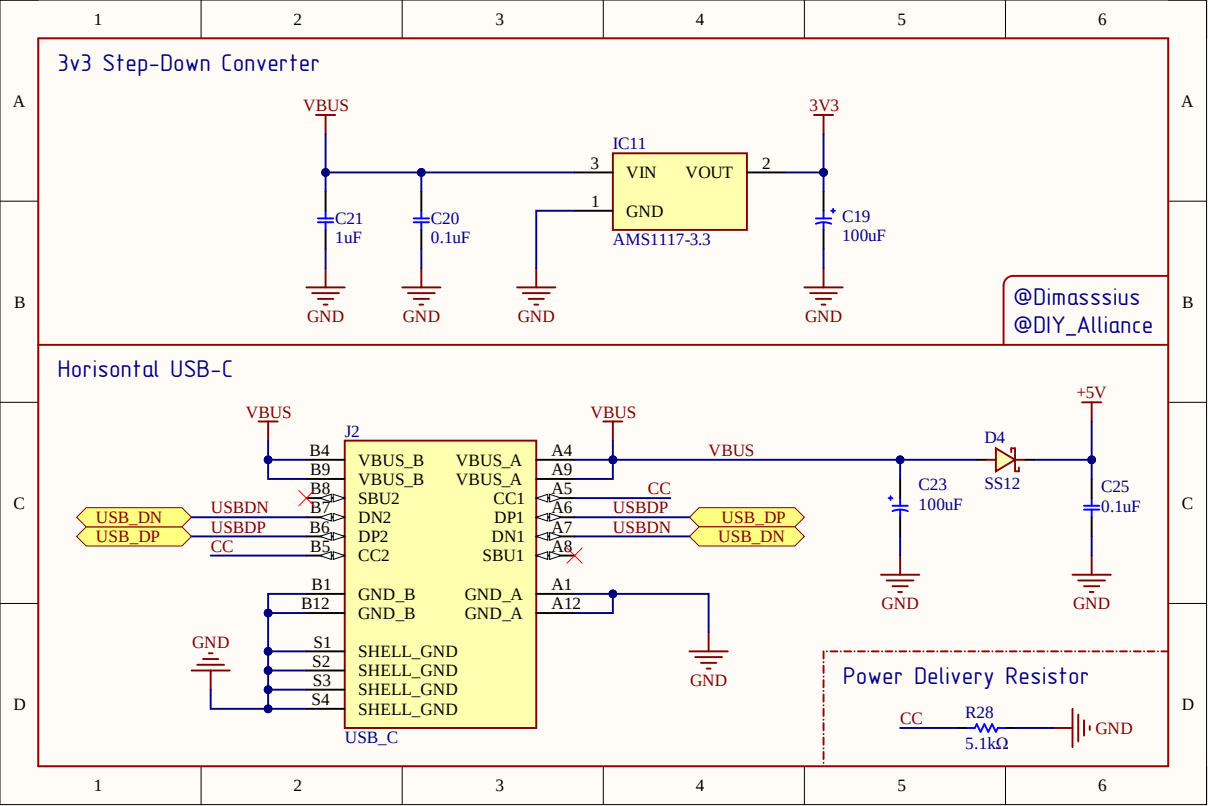


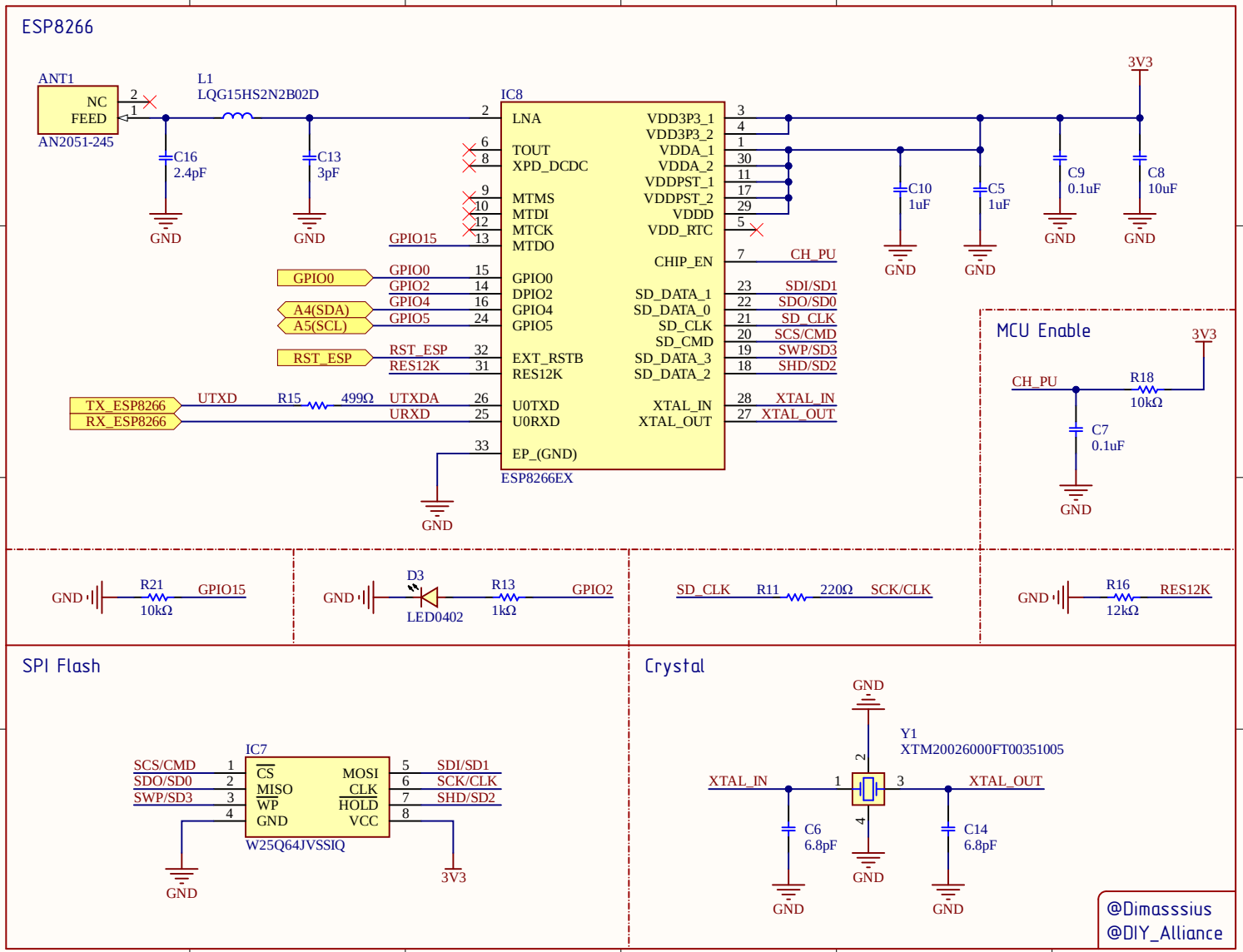
@Dimasssius
@DIY_Alliance

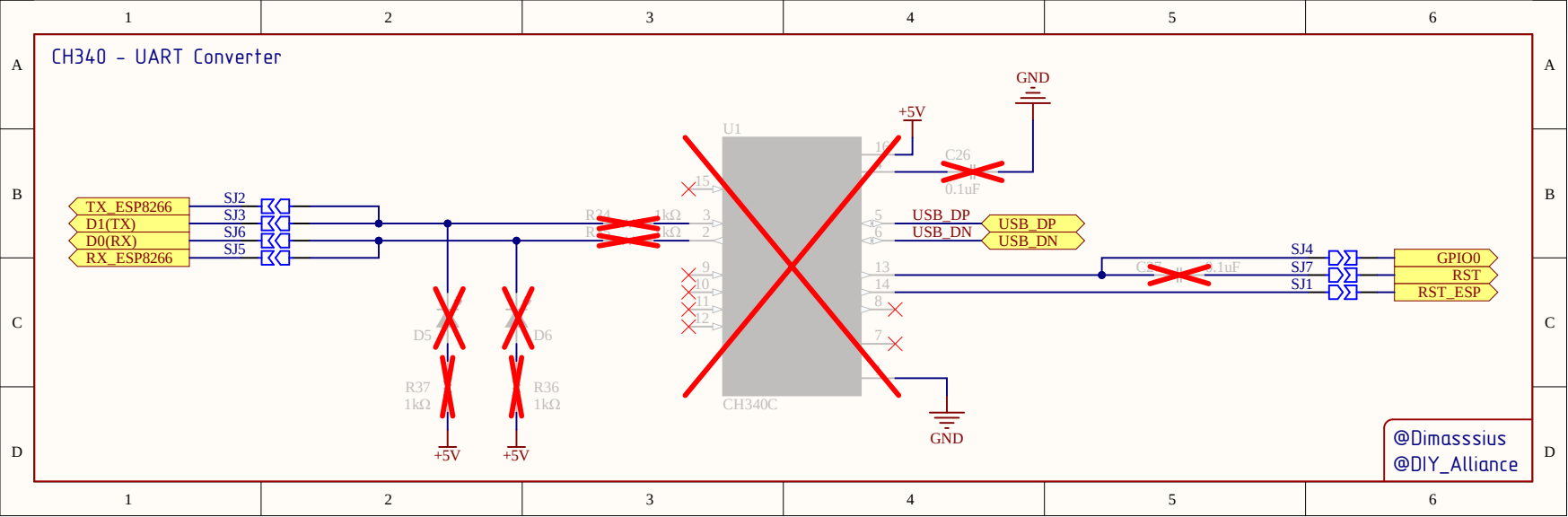



BME280
S6
+5V
A5(SCL)
A4(SDA)
BME280
GND
Infrared Receiever VS1838B / HS0038
D2
+5V
IR_RECIV
D8
TL1838
GND
Photoresistor
+5V
R34
10kΩ
R26
GL5528
A6
GND
Temperature Sensors
IC15
+5V
D3
DS18B20+
GND
IC16
+5V
R33
4.7kΩ
DHT11
GND
@Dimasssius
@DIY_Alliance


Buzzer
3V3
2
1
LS1
FUET-4020
2
1
Q4
MMBTA42
R23
100Ω
D7
GND
@Dimasssius
@DIY_Alliance


3v3 Step-Down Converter
VBUS
3V3
IC11
VIN
VOUT
GND
AMS1117-3.3
C21
1uF
C20
0.1uF
C19
100uF
@Dimasssius
@DIY_Alliance
Horisontal USB-C
J2
VBUS_B
VBUS_A
SBU2
CC1
DN2
DP1
DP2
DN1
CC2
SBU1
GND_B
GND_A
SHELL_GND
USB_C
USB_DN
USB_DP
USBDN
USBDP
CC
D4
SS12
C23
100uF
C25
0.1uF
+5V
Power Delivery Resistor
R28
5.1kΩ


ESP8266
ANT1
NC
FEED
AN2051-245
L1
LQG15HS2N2B02D
IC8
LNA
TOUT
XPD_DCDC
MTMS
MTDI
MTCK
MTDO
GPIO0
DPIO2
GPIO4
GPIO5
EXT_RSTB
RES12K
U0TXD
U0RXD
EP_(GND)
VDD3P3_1
VDD3P3_2
VDDA_1
VDDA_2
VDDPST_1
VDDPST_2
VDDD
VDD_RTC
CHIP_EN
SD_DATA_1
SD_DATA_0
SD_CLK
SD_CMD
SD_DATA_3
SD_DATA_2
XTAL_IN
XTAL_OUT
ESP8266EX
C16
2.4pF
C13
3pF
C10
1uF
C5
1uF
C9
0.1uF
C8
10uF
3V3
GND
GPIO15
GPIO0
GPIO2
GPIO4
GPIO5
RST_ESP
RES12K
CH_PU
SDI/SD1
SDO/SD0
SD_CLK
SCS/CMD
SWP/SD3
SHD/SD2
XTAL_IN
XTAL_OUT
A4(SDA)
A5(SCL)
TX_ESP8266
RX_ESP8266
UTXD
URXD
UTXDA
R15
499Ω
MCU Enable
R18
10kΩ
C7
0.1uF
R21
10kΩ
D3
LED0402
R13
1kΩ
R11
220Ω
SCK/CLK
R16
12kΩ
SPI Flash
IC7
SCS/CMD
SDO/SD0
SWP/SD3
CS
MISO
WP
GND
MOSI
CLK
HOLD
VCC
SDI/SD1
SCK/CLK
SHD/SD2
W25Q64JVSSIQ
Crystal
Y1
XTM20026000FT00351005
C6
6.8pF
C14
6.8pF
@Dimasssius
@DIY_Alliance


CH340 - UART Converter
TX_ESP8266
D1(TX)
D0(RX)
RX_ESP8266
SJ2
SJ3
SJ6
SJ5
R34
1kΩ
R35
1kΩ
D5
D6
R37
1kΩ
R36
1kΩ
+5V
+5V
U1
CH340C
GND
+5V
C26
0.1uF
GND
USB_DP
USB_DN
USB_DP
USB_DN
SJ4
SJ7
SJ1
GPIO0
RST
RST_ESP
@Dimasssius
@DIY_Alliance